A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 6: Security, Privacy and Data Integrity

Subtopic 6.2: Data Integrity

Concept 6.2.2: Data Verification Methods

Total Questions: 12

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_12 (Q1)

1 Draw **one** line from each verification method to indicate whether it is used during data transfer or data entry.

Verification Method	**Use**
Parity byte check	
Checksum	Data transfer
Visual check	Data entry
Parity block check	

[2]

Question 2 | Source: 9618_w25_qp_12 (Q6.c.i)

(c) A computer system uses even parity. The least significant (rightmost) bit of each byte is the parity bit.

(i) Complete the byte by writing the missing parity bit:

							parity bit ↓
0	1	0	1	1	1	0	

[1]

Question 3 | Source: 9618_w25_qp_12 (Q6.c.ii)

(c) A computer system uses even parity. The least significant (rightmost) bit of each byte is the parity bit.

(ii) The computer also uses parity block check. The parity block check uses even parity. Computer A transmits four bytes of data to computer B, followed by a parity byte. Computer B receives the following sequence of bytes.

								parity bit ↓
	1	0	1	1	0	1	1	1
	0	1	1	1	0	0	0	0
	0	0	0	1	1	0	1	1
	0	1	1	1	0	1	0	0
parity byte →	1	0	1	0	0	0	0	0

Following transmission, one of the four bytes of data has an error in one of the bits.

Circle the bit that has been altered during the data transfer. [1]

Question 5 | Source: 9618_s25_qp_11 (Q7.c)

7 A banker stores personal data on their work computer.

(c) The data that is transferred can also be verified using a checksum.

Explain how data can be verified using a checksum.

....

....

....

....

....

.... [3]

Question 6 | Source: 9618_s25_qp_12 (Q5.c.ii)

5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

```
USER(Username, Password, DateOfBirth)

CHARACTER(CharacterName, CharacterID, Username, Level, Money)

ITEM(ItemName, MinimumLevel, Cost)

CHARACTER_ITEM(CharacterID, ItemName)
```

(c) A Database Management System (DBMS) provides data security.

(ii) The DBMS also supports data integrity.

Give **two** ways that a DBMS can support data integrity.

1 ....

....

2 ....

....

[2]

Question 7 | Source: 9618_s24_qp_11 (Q5.b)

5 A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

(b) The bank wants to protect the integrity of its data while transferring the data to other banks. Parity check is one example of data verification.

Complete the description of parity check when Computer A is transmitting data to Computer B.

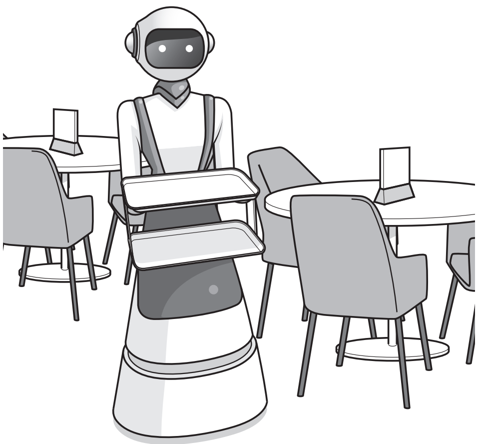
Computer A and Computer B agree on whether to use .... parity. Computer A divides the data into groups of .... The number of 1s in each group is counted. If the agreed parity is .... and the group has an even number of 1s, a parity bit of 1 is appended, otherwise a parity bit of 0 is appended.

In a parity .... check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity ....

[5]

Question 8 | Source: 9618_s24_qp_13 (Q7.f.i)

7 Robots are used to serve food and drink to customers at a restaurant.



(f) The data from the robots is transmitted to a central computer using a wireless connection.

(i) Complete the table by identifying **and** describing **two** methods of data verification that can be used during data transfer.

	Method	Description
1	....	....
2	....	....

[4]

Question 9 | Source: 9618_w23_qp_13 (Q3.c)

3 A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

(c) An example of a tutor ID is NK16C6.

An administrative officer enters the tutor ID into the `TUTOR` table.

Explain how data verification can be used when the tutor ID is entered.

....

....

....

....

....

....

....

.... [4]

Question 10 | Source: 9618_w23_qp_13 (Q8.a)

8 **(a)** Data verification is one method of protecting the integrity of data.

Describe **one** other method of protecting the integrity of data.

....

....

....

.... [2]

Question 11 | Source: 9618_s23_qp_13 (Q4.d.ii)

4 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(d) The data is validated and verified when it is entered into the database.

(ii) Describe **two** ways that the car registration number can be verified when it is entered into the database.

1 ....

....

2 ....

....
[2]

Question 12 | Source: 9618_w22_qp_12 (Q4.b)

(b) A checksum can be used to detect errors during data transmission.

Describe how a checksum is used.

....

....

....

....

....

.... [3]

Question 13 | Source: 9618_w21_qp_11 (Q2.b.ii)

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(b) Xanthe uses both data validation and data verification when entering data on her computer.

(ii) Describe how data verification helps to protect the integrity of the data. Give an example in your answer.

Description ....

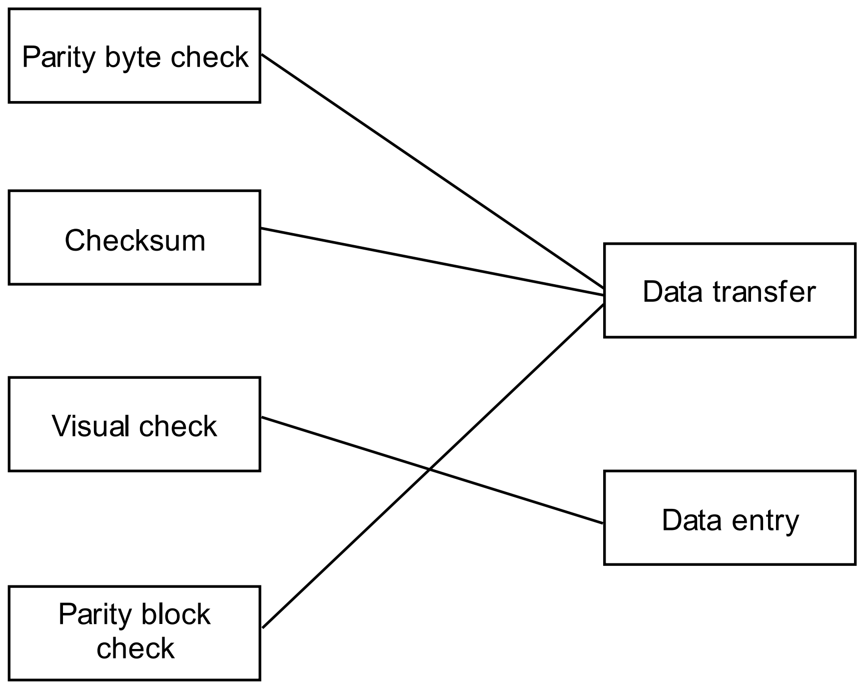
....

....

Example ....
[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q1)

Question	Answer	Marks
1	**1 mark** for 2 or 3 correctly connected verification method boxes, **2 marks** for all 4 correct **Verification Method** **Use** 	**2**

Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q6.c.i)

Question	Answer	Marks
6(c)(i)	**1 mark** for correct answer	**1**

parity bit ↓

0	1	0	1	1	1	0	**0**

Mark Scheme for Question 3 | Source: 9618_w25_ms_12 (Q6.c.ii)

6(c)(ii)	**1 mark** for correct answer (table below) [1]	**1**

								parity bit ↓
	1	0	1	1	0	1	1	1
	0	1	1	1	0	0	0	0
	0	0	0	1	1	0	1	1
	0	1	1	1	0	1	0	0
parity byte→	1	0	1	0	0	0	0	0

Mark Scheme for Question 4 | Source: 9618_w25_ms_13 (Q7.c)

7(c)	**1 mark** for the name **2 marks** for the matching description, **max 3** marks • Parity Byte Check • a parity bit is added to each byte to make the number of 1s match the parity, odd or even • each byte can be checked on receipt and request to be resent if the byte does not match parity • Parity Block Check • a bit is added to each byte, but a parity byte is also set for each block • the location of an error can be found using vertical and horizontal parity • Checksum • A calculation is made from the data and transmitted with the data • The receiver performs the same calculation and compares with received checksum to see if they match	**3**

Mark Scheme for Question 5 | Source: 9618_s25_ms_11 (Q7.c)

7(c)	**1 mark** each to **max 3** • The data is put through an algorithm to create a checksum value • The data and checksum are sent to the receiver • The receiver performs the same algorithm on the data • if both checksums match the data is verified	**3**

Mark Scheme for Question 6 | Source: 9618_s25_ms_12 (Q5.c.ii)

5(c)(ii)	**1 mark** each to **max 2** e.g. • Validation • Enforce referential integrity • Cascade update / delete • Ensuring the database is normalised	**2**

Mark Scheme for Question 7 | Source: 9618_s24_ms_11 (Q5.b)

5(b)	**1 mark** for each correctly completed term: • odd or even • 7-bits • odd • block • byte Computer A and Computer B agree on whether to use **odd or even** parity. Computer A divides the data into groups of **7-bits**. The number of 1s in each group is counted. If the agreed parity is **odd** and the group has an even number of 1s a parity bit of 1 is appended, otherwise a parity bit of 0 is appended. In a parity **block** check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity **byte**.	**5**

Mark Scheme for Question 8 | Source: 9618_s24_ms_13 (Q7.f.i)

7(f)(i)

1 mark for each correct method and
1 mark for corresponding description to **max 4**:

4

Method	Description
Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.
Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.
Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.

Mark Scheme for Question 9 | Source: 9618_w23_ms_13 (Q3.c)

3(c)	**1 mark** for each bullet point. • The administrator completes a visual check / checks by eye • …that the tutor identifier input matches the tutor identifier on the **original** document • Double entry check // the administrator (or a second person) enters the number a second time • …and the **system** compares it with the first entry	**4**

Mark Scheme for Question 10 | Source: 9618_w23_ms_13 (Q8.a)

8(a)	**1 mark** for each bullet point (**max 2**) • Validation // a validation method named or described • …protects the data by ensuring that the data is reasonable / sensible and within specified bounds	**2**

Mark Scheme for Question 11 | Source: 9618_s23_ms_13 (Q4.d.ii)

4(d)(ii)	**1 mark** each • Visual check: **Manually** compare the registration number entered with the source document • Double entry: Enter the registration number twice and **the computer compares** to check they are the same	**2**

Mark Scheme for Question 12 | Source: 9618_w22_ms_12 (Q4.b)

4(b)	**1 mark** for each bullet point **(max 3)**: • checksum value is calculated **from the data** before transmission // correct description of a checksum algorithm • ... this calculated value is transmitted with the data • **receiving** computer recalculates the checksum from the received data • if the checksum **received** and **calculated** match, no error has occurred // if the checksum **received** and **calculated** do not match, an error has occurred	**3**

Mark Scheme for Question 13 | Source: 9618_w21_ms_11 (Q2.b.ii)

2(b)(ii)	**1 mark** per bullet point • verification checks that data is the same as the original • by example e.g. double entry	**2**